Phytoliths as climate clues

Tiny silica plant structures from soil could track temperature changes

By Sid Perkins

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TORONTO — Analyses of tiny silica structures that form in the leaves and wood of many plants can yield information about the temperature in which the plants grew, a new study suggests.

Phytoliths — which means "plant stones" — are minuscule, often distinctly shaped crystals of silica that form in vegetation as a plant grows. And they're long-lasting: Paleontologists have used phytoliths trapped in fossils to infer the diet of some dinosaurs (SN:10/20/01, p. 248). Now, scientists might be able to use phytoliths from long-dead plants unearthed from soil as paleothermometers, Zhenzhen Huang, an isotope chemist at the University of Western Ontario in London, Canada, reported May 26 at an American Geophysical Union meeting.

Huang and her colleagues grew cattails and horsetails, types of marsh plants, in a climate-controlled chamber in the lab. During the team's eight-month-long tests, environmental conditions were held constant — the temperature for one set of plants was 15° Celsius, and other groups were grown at 20°C, 25°C and 30°C.

As data from previous field tests had hinted, the ratio of oxygen-16 and oxygen-18 isotopes in the plant-produced silica varied according to the temperature in which the plants grew, Huang said. On average, the higher the temperature, the lower the ratio of oxygen-18 to oxygen-16.

Previous studies have shown that phytoliths from fallen and decomposed vegetation can persist in soil for up to 300,000 years, Huang said. So, she speculates, oxygen-isotope data from phytoliths gleaned from soil — if combined with the results of carbon-dating organic material from the same soil sample — could provide data about Earth's climate during the last three ice ages and interglacial periods.

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CITATIONS

Huang, Z., E.A. Webb, and F.J. Longstaffe. 2009. Oxygen isotope of phytoliths in modern wetland plants and the application to paleoclimate reconstruction (Presentation CG21A-16). 2009 Joint Assembly. May 24-27. Toronto, Ontario. [Go to] (Enter "phytoliths" in the search box)